QUASI-AFFINENESS CHARACTERIZED BY THE FACTORIZATION THROUGH THE RELATIVE SPECTRUM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the canonical factorization named in a remark of this chapter; the remark is a navigation pointer and the proof works with the factorization as displayed.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Quasi-affineness is local on the base. Lemma lemma-quasi-affine-local, used at proof lines 4281-4283. Allows the reduction to the case of a scheme base.
- The base change formula for pushforwards of structure sheaves along an etale morphism. Properties of Spaces, Lemma spaces-properties-lemma-pushforward-etale-base-change-modules, used at proof lines 4283-4286. Identifies the restriction of f_*O_Y with the pushforward along the base change.
- Formation of the relative spectrum commutes with base change. Lemma lemma-affine-equivalence-algebras, used at proof lines 4286-4289. Makes the factorization compatible with the reduction.
- The scheme case of the characterization of quasi-affine morphisms. Morphisms, Lemma morphisms-lemma-characterize-quasi-affine, used at proof lines 4290-4293. Is the input applied once X and Y are schemes.

Source reference. Original lemma 081V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10230–10230. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5117–5117. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01SM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2343–2343. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 081X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10232–10232. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4729–4729. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. A quasi-compact and quasi-separated morphism of algebraic spaces $f : Y \to X$ is quasi-affine if and only if the canonical factorization $Y \to \underline{\mathrm{Spec}}_X(f_*\mathcal{O}_Y)$ (Remark 081X) is an open immersion.*

Proof. Let $U \to X$ be a surjective morphism where U is a scheme. Since we may check whether f is quasi-affine after base change to U (Lemma 03WM), since $f_*\mathcal{O}_Y|_V$ is equal to $(Y \times_X U \to U)_*\mathcal{O}_{Y \times_X U}$ (Properties of Spaces, Lemma 03LX), and since formation of relative spectrum commutes with base change (Lemma 081V), we see that the assertion reduces to the case that X is a scheme. If X is a scheme and either f is quasi-affine or $Y \to \underline{\mathrm{Spec}}_X(f_*\mathcal{O}_Y)$ is an open immersion, then Y is a scheme as well. Thus we reduce to Morphisms, Lemma 01SM. □